A TOOL FOR LECTURE ASSESSMENT AS PER OBE POLICY

SUBMITTED IN PARTIAL FULFILLMENT FOR THE REQUIREMENT OF THE AWARD OF DEGREE OF

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE

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Session 2025-26

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May 2026

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We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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This is to certify that Project Report entitled "A tool for lecture assessment as per OBE policy" which is submitted by Pallavi Sharma, Saksham in partial fulfillment of the requirement for the award of degree B. Tech. in Department of Computer Science of Dr. A.P.J. Abdul Kalam Technical University, Lucknow is a record of the candidates own work carried out by them under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

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ACKNOWLEDGEMENT

It gives us a great sense of pleasure to present the report of the B. Tech Project undertaken during B.Tech. Final Year. We owe special debt of gratitude to Professor Dheeraj Pandey, Department of Computer Science, KIET, Ghaziabad, for his constant support and guidance throughout the course of our work. His sincerity, thoroughness and perseverance have been a constant source of inspiration for us. It is only his cognizant efforts that our endeavors have seen light of the day.

We also take the opportunity to acknowledge the contribution of Dr. Ajay Kr Shrivastava, Dean of the Department of Computer Science, KIET, Ghaziabad, for his full support and assistance during the development of the project. We also do not like to miss the opportunity to acknowledge the contribution of all the faculty members of the department for their kind assistance and cooperation during the development of our project.

Last but not the least, we acknowledge our friends for their contribution in the completion of the project.

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ABSTRACT

In a modern educational context, engagement and the development of better learning outcomes rely heavily on the integration of technology. This project involves designing a dedicated hardware device to be used in assisting teachers to run MCQ assessments among students with the intent of increasing their participation while reducing possible mobile device distractions.

The device will offer a product that enables students to interact and respond to MCQ in a manner that is interactive and real-time, helping educators to measure their level of comprehension appropriately. The device will, by not requiring a smartphone or a tablet for examinations, offer a product that encourages a conducive learning culture by lessening the incidence of distracting behaviour.

The device is expected to have some important features such as user-friendliness, instant feedback, and analytics of student performance over time. The device is also expected to allow teachers to create, modify, and distribute question sets best suited to the course, and the device will allow different types of questions and offer instant results.

This particular project can also be categorised as being in line with the principles advocated by Outcome-Based Education (OBE), where emphasis is always placed on the outcome. By ensuring that teachers will now have a sure way to encourage active student participation during MCQ assessment, it can therefore be seen that this project aims to create a more interactive and conducive environment for teaching and learning. By ensuring the success of this project, it is also guaranteed that students will be more encouraged to be more productive.

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LIST OF ABBREVIATIONS

AI – Artificial Intelligence

CAS – Classroom Assessment System

EdTech – Educational Technology

GUI – Graphical User Interface

ICT – Information and Communication Technology

IoT – Internet of Things

JWT – JSON Web Token

LMS – Learning Management System

MCQ – Multiple Choice Question

NFC – Near Field Communication

NFR – Non-Functional Requirement

OBE – Outcome-Based Education

RFID – Radio Frequency Identification

SDLC – Software Development Life Cycle

SRS – Student Response System

SSL – Secure Sockets Layer

TLS – Transport Layer Security

UID – Unique Identifier

SDG MAPPING

The development of the proposed project meets the UN Sustainable Development Goals (SDGs), ensuring that the final solution is both technologically advanced and contributes to a sustainable society. Combining new classroom assessment technology with education goals, the project supports global initiatives to advance the quality of education as well as stimulating innovation and promoting inclusive learning environments. The project supports the following SDGs, as described below.

SDG 4: Quality Education

The proposed Hardware-Based MCQ Classroom Assessment System primarily aligns with Sustainable Development Goal 4: Quality Education, which focuses on ensuring inclusive and equitable quality education and promoting lifelong learning opportunities for all. The system improves the classroom interaction as it allows real-time, interactive MCQ-based assessment without using the mobile phone, which reduces distraction to the maximum. The project is based on the principles of the Outcome-Based Education (OBE) that enables teachers to map the assessments to course outcomes and learning objectives. The performance tracking and real-time analytics functions serve as the means to support the use of data-based teaching approaches to enable educators recognize the learning gaps to enhance the teaching approaches. In that way, the project will improve the quality of education system, its transparency, and effectiveness.

SDG 9 : Industry, Innovation and Infrastructure

Sustainable Development Goal 9: Industry, Innovation and Infrastructure also supports the project in that it focuses on creating resilient infrastructure and stimulating innovation. The system combines the hardware embedded with IoT and a web-based solution, including

React.js, Node.js, and MySQL, and it encourages technology development in the educational industry. The project promotes the creation of smart classroom infrastructure by launching a special inexpensive hardware used to conduct classroom-based assessments. This is a contemporary form of evaluation that substitutes the conventional paper-based and smartphone-designed systems with a secure and suitable technological system, which leads to the transformation of the educational institutions to the digital world.

SDG 10: Reduced Inequalities

The suggested system indirectly serves Sustainable Development Goal 10: Reduced Inequalities because every student would participate in the system and be judged equally and without discrimination. Being a standardized hardware that is used by each and every student to post responses means that the system removes bias and provides equal assessment conditions. The online assessment system is fair in marking and assessment as it offers clear and precise performance analysis. The project helps to achieve the overall goal of decreasing disparities in access to quality education by facilitating the active inclusion and equal opportunity in the classroom.

CHAPTER 1
INTRODUCTION

1.1 Introduction

Nowadays, mobile devices are ubiquitous in schools, and they may be quite useful and a major nuisance at the same time. I just mean that they are used by students to look things up or whatever, but in the classroom they only end up causing distractions such as when they are doing quizzes, people are taking looks at their phones. The teachers struggle to get things straight and it disrupts the ability of the children to listen or even comprehend.

It is a project of fixing that with our new hardware thing we made. It is simply a device that simply answers multiple-choice questions without the use of a phone whatsoever. It allows students to utilize it during classes, and it ensures they do not get distracted, thus perhaps they are more engaged. It is to make it more productive in class, you know, with all the breaks.

What is unique is that it links it to software that evaluates answers as they are being done. The teachers receive instant feedback on individual student performance and it also keeps track of the performance to identify where one person may be lagging behind. In that manner, they can modify lessons or something to make it better. It coincides with OBE that is all about gauging what students learn and refining it.

The integration of the software and the device such as this would actually reduce distraction during assessments. It appears to lean towards the direction of a more interactive and results-driven setup in class which is not necessarily working perfectly yet.

1.2 Project Category

The suggested initiative is known as A Tool to Lecture Assessment according to OBE policy and belongs to the category of the IoT-enabled Educational Technology (EdTech) solution that combines hardware and software. It is an interdisciplinary initiative that integrates embedded systems, web development, database management and real time data processing. It is a system consisting of the creation and deployment of a special hardware apparatus to be used by students to answer multiple-choice questions that will connect to a central storage and analysis server.

Technologically, this project will be classified as a full-stack application with the built-in IoT infrastructure, with the frontend code developed with React.js, the backend code written with Node.js, and MySQL to manage databases. It is a smart classroom solution because the system facilitates real-time communication between the hardware devices and the web-based dashboard.

Regarding the area of application, the project falls into the industry of Smart Education Systems and aids in the implementation of the Outcome-Based Education (OBE) practices. It aims at enhancing classroom activities, assessment precision and performance statistics by a safe and scalable technology framework.

1.3 Objectives

This project is aimed at developing and creating a specialized hardware device, which is integrated with a corresponding software system to support Multiple Choice Question (MCQ)-based tests in the classroom. Specifically, this device aims at reducing distractions by removing the use of mobile phones that usually result in the lack of student focus and engagement. As per the concepts of the Outcome-Based Education (OBE), it is supposed to develop an instrument that will enable interactive, real-time engagement during testing in order to enable educators to better gauge the knowledge of students. The hardware will give

the students a specific platform to answer MCQs in classes so that they are not distracted by notifications, social media, and other activities that are not educational.

Integrated software will track and analyze the responses of every student, and provide immediate response to the teachers. Such live data will allow teachers to determine the areas of learning deficiency, assess the performance of students to the learning outcomes and modify the instructional processes accordingly. The project will attempt to make the learning environment more active, with more output-based, through the active involvement of the participants and the provision of real-time feedback. Finally, the project would assist teachers in enhancing classroom concentration, the quality of learning and offering credible performance reporting that conforms to the objectives of OBE which would see students attain the desired educational content.

1.4 Structure of Report

The report is subdivided into a few chapters that are devoted to a particular step of the project development process.

Chapter 1: Introduction

The chapter is the introduction to the project that revealed the background, the category of the project, the objectives, and the general intent of the study. It describes how a secure and distraction-free system of classroom assessment is necessary and gives a summary of the proposed solution.

Chapter 2: Literature Review

The chapter is a review of the research on digital assessment tool quizzes and interactive classroom response system that is currently available. It reveals the past research, gaps in research, and develops the problem of the current research in this project.

Chapter 3: Proposed System

In this chapter, the proposed RFID/NFC-based lecture evaluation gadget is discussed in detail. It discusses the system architecture, how it works, and the distinguishing characteristics that make it different to the available internet-based solutions.

Chapter 4: Requirement Analysis and System Specification

This chapter presents the feasibility study, software requirements, system specifications, design diagrams, and database structure. It defines the functional and non-functional requirements of the system.

Chapter 5: Implementation

This chapter explains the tools and technologies used in the development of the project. It describes the hardware setup, system integration, and implementation process.

Chapter 6: Testing and Maintenance

This chapter discusses the testing techniques and validation methods used to ensure system reliability, accuracy, and performance. It also outlines maintenance considerations.

Chapter 7: Results and Discussion

This chapter presents the system outputs, module descriptions, performance observations, and key findings derived from implementation and testing.

Chapter 8: Conclusion and Future Scope

This chapter summarizes the overall outcomes of the project and discusses possible future enhancements and improvements to the proposed system.

CHAPTER 2
LITERATURE REVIEW

2.1 Literature Review

The introduction of assessment tools based on digital quizzes has greatly revolutionized approaches to classroom practice in the modern environment contributing to increased student interest, active involvement, and making the real-time comparison of the learning outcomes possible. Recent studies reiterate that interactive quiz applications will give instant feedback and teachers can dynamically modify their models of teaching with consideration of the Outcome Based Education (OBE) rules. The efficacy, participation, and educational effect of generally utilized Internet-based assessment instruments like Kahoot!, Quizizz, Mentimeter, Socrative, and ClassPoint have been explored in a few studies.

Amalia et al. (2024) and Wang et al. (2016) remark in their works that gamified interfaces like Kahoot! and Quizizz offer a considerable enhancement in terms of analytical reasoning and motivation in students, which, in turn, can be attributed to the argument of the researcher that the audience should cultivate healthy competition, leaderboards, and instant feedback with other participants to become more motivated learners. The gamification strategy would boost engagement and motivation, especially classroom education. On the same note, Zainuddin et al. (2020) noted that gamified e-quizzes have positive effects on academic outcomes and student engagement when they are used as a form of a formative assessment. Nevertheless, such platforms are usually very dependent on the internet and smart devices that can cause distractions and technical interruptions.

As also studied by Antonenko and Izmailova (2023) and Armstrong et al. (2024), Mentimeter and ClassPoint concentrate on interactive presentations and real-time texting in order to promote collaborative and anonymous involvement. Such tools come in handy especially in motivating the silent students who are shy enough not to be judgmental. Socrative, which is

mentioned in the works by Fonseca et al. (2024) and Ingalls (2020), allows various types of questions and in-depth performance statistics, which is why it is applicable in both formative and summative evaluations. Students in formal educational settings have found themselves liking Socrative since it is easy and it helps to monitor performance deliverables.

2.2 Research Gap

A thorough review of existing literature indicates several key gaps in currently available digital classroom assessment technologies. A primary limitation of most interactive quiz platforms is that they are very reliant on reliable internet connectivity. This severely limits their dependability within an educational setting that does not offer consistent or no access to a network. An interruption to the internet connection will either stop assessments from occurring or be able to collect responses inaccurately.

A second area of concern is that many of the interactive quiz platforms require use of students' own mobile phones or other personal digital devices to participate. Although this increases accessibility to assessments, it also increases the likelihood that a student will be distracted by non-academic activities (e.g., social media) during a lecture. In addition, the availability of suitable devices varies across students; thus, some students are at a disadvantage when participating in assessments and therefore experience inequity in their assessments.

Several privacy/data security risks exist in the use of many digital assessment platforms. These platforms typically utilize cloud-based storage services and third-party servers to store and manage the data collected from students, thereby increasing concerns regarding student data confidentiality and potential unauthorized access to student data. Furthermore, institutions often lack control over how student data is collected, maintained, and used in the use of these platforms.

In addition to the previously mentioned limitations, many of the online assessment platforms are subscription/contract based (i.e., recurring expense) and/or require institutional purchasing

of hardware and software infrastructure to support use of the platform. As such, these recurring costs and required expenditures are likely to limit long-term institutional adoption of these systems.

Research indicates that the application of gamification techniques enhances engagement, however, few studies have investigated the development of secure, offline, and affordable, hardware-based solutions for classroom assessment purposes. There exists a significant need for a system that eliminates the need for internet connectivity, minimizes digital distractions, maintains confidentiality, and is affordable, while still providing for effective classroom engagement.

2.3 Problem Formulation

The primary problem to be addressed in this project as described by the research gaps are issues with the design of an affordable, dependable and secure Classroom Assessment System (CAS) that is independent from internet connection while providing for both the integrity and privacy of the responses of students.

Even the available digital assessment tools are effective in improving engagement and are to a great extent reliant on the presence of Information and Communication Technology (ICT) infrastructure and individual smart devices, which restricts their use in controlled academic settings.

Conventional online quiz systems that are used in Multiple Choice Question (MCQ) assessment are very dependent on smartphones, tablets, and having constant access to the internet. This dependency would usually lead to student distraction and high chances of malpractice coupled with technical interruptions due to unstable network connection. Use of personal digital devices in the assessment activities could cause multitasking and off-task behavior thus lowering the effectiveness of lectures. Additionally, the unequal access to digital materials introduces unequal access to the learners and influences the aspects of fairness and inclusiveness in assessment.

The other issue of concern is associated with the privacy and security of data. The majority of online Student Response System (SRS) save the performance information on third-party clouds that creates a threat of unauthorized access, abuse of data and inability of institutions to control sensitive academic information. These problems are inconsistent with the values of safe academic assessment and can limit the use of such platforms in institutes.

The suggested system aims to eliminate such constraints by introducing an RFID/NFC-based offline lecture evaluation device which will remove the reliance on the availability of the internet and mobile devices. This system allows students to send MCQ feedback through RFID/NFC tags and the retrieved data is stored in the system locally in the database. This guarantees data integrity, confidentiality and continuous flow of assessment in controlled settings of the classroom.

Thus, the formulation of the problem focuses on the development of a distraction-free, privacy-focused, and cost-efficient assessment system, which corresponds to the principle of the Outcome-Based Education (OBE), covering the effective measurement of the learning outcomes without using third-party digital infrastructure.

CHAPTER 3
PROPOSED SYSTEM

3.1 Proposed System

The proposed system is a Classroom Assessment System (CAS) that will be based on RFID/NFC to achieve MCQ-based tests in secure and offline, and distraction-free environment. The system will assist in facilitating the Outcome Based Education (OBE) through real time evaluation of student comprehension without the use of smart phones or even connection to the internet. This solution is hardware-based and unlike the conventional online Student Response System (SRS) operates without the external digital infrastructure.

The device enables students to give answers on RFID/NFC tags with option A, B, C, and D. An RFID reader, which is attached to the system of the teacher, records these replies and the answer is simply stored in a local database, where it can be evaluated and analyzed further.

3.1.1 System Architecture

The architecture of the proposed system consists of three major components: RFID/NFC tags (student input unit), RFID reader module (data acquisition unit), and a processing and storage unit (teacher interface with local database).

Each student is provided with four RFID/NFC tags representing MCQ options. When a student selects an answer, the corresponding tag is activated and detected by the RFID reader. The reader transmits the tag ID to the processing unit, where the response is recorded and mapped to the respective question.

The system operates in an offline mode, ensuring uninterrupted performance without dependency on Wi-Fi, mobile data, or cloud-based servers. All response data is stored locally, thereby maintaining confidentiality and integrity.

3.1.2 Working Mechanism

The functional mechanism of the system relies on the RFID/NFC communication technology. Every RFID has a unique identifier, known as the UID, which is embedded in a microchip. When the RFID is brought close to the RFID reader, it is activated using electromagnetic induction.

In a lecture setting, the lecturer delivers an MCQ either orally or using the traditional board method. The students respond to the MCQs by turning on the RFID corresponding to their answer. The RFID reader reads the responses in real time.

Once the responses are collected, the lecturer enters the correct answer to the system, which automatically calculates the performance of the students. The system provides instant results, including the total number of responses, correct responses, and performance distribution.

3.1.3 Hardware Components

The proposed system is made out of cheap and readily sourced hardware. These consist of RFID/NFC student response input, RFID reader module to take in data, a microcontroller or processing unit to process the inputs and a local storage unit or database to store data on responses.

The hardware implementation makes sure that the power consumption is minimal and the operation costs are low. The system does not need constant internet connecting, and therefore will use less energy as opposed to smartphone-based assessment platforms.

3.1.4 Data Processing and Storage

All responses from the students are processed locally. The tag IDs obtained are then matched with the predefined options for the MCQs and are stored in a database. The local storage system is secure and does not pose any risks associated with cloud storage.

The system allows for automated marking by matching the responses with the instructor's entry of the correct answer. The results can be presented in the form of summary statistics, which can help in decision-making based on OBE.

3.1.5 Advantages of the Proposed System

The suggested CAS based on RFID/NFC does not have significant constraints as found in the current digital assessment tools. It eliminates internet addiction, reduces distraction among students, assures the privacy of information, and decreases operation expenses. The system is also scalable and can be deployed easily as well as being applicable to institutions that have low ICT infrastructure.

It is a secure and efficient alternative to the traditional online quiz platforms offering active involvement of the classroom setting by means of combining hardware simplicity with the capability of real-time evaluation.

3.2 Unique Features of The System

The proposed RFID/NFC-based Classroom Assessment System (CAS) will fill the gaps of current digital quiz systems with the aspects of simplicity, security and efficiency into a single hardware-driven system. There is absolutely no online presence of the system and this is one of the most conspicuous parts of the system. The proposed system does not rely on network infrastructure unlike the traditional Student Response Systems (SRS) that require constant internet connection and cloud-based servers to operate. All the responses of the students are

recorded by the RFID reader and processed with the Arduino board and stored in the system locally. This will guarantee continuous functioning even in the classrooms where there is no/poor internet connectivity. Consequently, the quality of assessments becomes much better, and the danger of being technically interrupted throughout lectures is reduced.

The other feature of the system is the capability to establish a distraction free academic environment. In the case of a conventional online quiz-based assessment, students access smartphones or other digital-based devices which can expose them to social media, messaging apps, gaming sites or other non-academic distractions. This tends to decrease concentration and influence learning. The proposed system will eliminate this type of device rather than use their money in purchasing it and adding NFC tags which are aimed at just submitting academic answers. This enclosed environment will ensure that the students do not get to be interrupted by external digitization in regard to remaining involved in the lecturing and evaluation process.

The system design is concerned with the data confidentiality and integrity. Unauthorized access, data leakage and privacy invasion risks have been reduced to the minimal because all the data is stored locally and is not transmitted to the third party cloud servers. NFC tags contain a unique identifier (UID) with the help of which they can be correctly mapped when relied on the responses of particular students or anonymous mode of participation as necessary. The localized data storage system will boost the institutional control of the academic records and is ethical according to the data protection standards.

Cost-effectiveness would be another important advantage of the proposed solution. Majority of online assessment vendors operate on subscription or on licensing model such that when a user requires more advanced features such as analytics, reporting or AI-driven question generation, they must pay on a regular basis. In contrast, RFID/NFC-based CAS suggested in this paper presupposes one hardware investment. Arduino board, RFID reader, and NFC tags are not costly and can be easily sourced and they do not require a lot of maintenance. This

makes the system very suitable when the institutions to be addressed have a tight budget or low ICT resources.

The system also supports real-time acquisition of responses, and automated evaluation. Upon the student selecting an NFC tag as an MCQ option, the RFID reader will directly pick up the tag and relay the UID to the Arduino to work with. The system and automation of comparison and performance analysis can be created with the teacher in the system inputting the right answer. This type of feedback system supports the concepts and values of the Outcome-based education (OBE) since it enables teachers to gauge the learning outcomes during the very lecture and modify the instruction.

Scalability and flexibility also enhance the applicability of the system. The module design allows the simplicity of the addition of more NFC tags to serve bigger classes. Further expansion Future expansions The Arduino platform can be extended to either a graphical user interface (GUI) or performance dashboards or statistical analysis modules. This will mean that the system will be in a position of developing as the requirements of the institution are required without significant structural adjustments.

Another characteristic worth mentioning is energy efficiency. NFC tags are passive and do not have any internal power source, the devices have an electromagnetic field that is produced and sent by the RFID reader. The Arduino board has a low power consumption and therefore the whole system is energy efficient as compared to smart phones based assessment techniques which have high power consumption in terms of battery consumption and network usage.

CHAPTER 4
REQUIREMENT ANALYSIS AND SYSTEM SPECIFICATION

4.1 Feasibility Study

The feasibility study examines the viability of the proposed RFID/NFC-based Classroom Assessment System (CAS) by determining whether it is possible to develop and implement the suggested system within feasible contexts. It examines the system in three views which are Technical Feasibility, Economic Feasibility, and Operational Feasibility. This analysis will make sure that the system is feasible, viable as well as manageable to be used in classroom and in the framework of Outcome-Based Education (OBE).

4.1.1 Technical Feasibility

Technical feasibility will be used to define the availability and capability of the necessary technology, hardware, and software resources to support the proposed system. RFID reader, the Arduino board and the NFC tags are all common and compatible parts. The commutation of the RFID reader and the Arduino is possible with the help of the traditional communication protocols (SPI/UART). Arduino is programmed in the Arduino IDE that is easy to use and documented. The system is also offline, so it does not need to have complicated ICT infrastructure like servers, routers, and cloud services. The local data storage system will not rely on internet connectivity to achieve a stable performance. Scalability and future enhancement are also possible by the use of the modular architecture.

Table 4.1 Technical Feasibility Analysis

Parameter	Existing Online Systems	Proposed RFID/NFC-Based CAS
Internet Requirement	Mandatory	Not Required
Hardware Complexity	High (Smartphones, Servers)	Low (RFID + Arduino + NFC)
System Integration	Cloud-Based	Local Processing
Scalability	Dependent on Infrastructure	Easily Expandable
Reliability	Affected by Network Issues	Stable Offline Operation

4.1.2 Economic Feasibility

Economic feasibility is used to determine whether the system is cost effective to be developed and maintained on a realistic budget. The suggested CAS will need a single investment in hardware devices such as the RFID reader, Arduino board and NFC tags. These are cheap parts and easily bought in the market. In contrast to the subscription-based quiz sites, the offered system is not associated with the recurring licensing charge or cloud hosting expenses. The simple hardware configuration and low power consumption mean that the maintenance costs are low. As such, the system is economically viable to institutions that have minimal ICT funds.

Table 4.2 Economical Feasibility Analysis

Cost Factor	Online Quiz Platforms	Proposed RFID/NFC CAS
Initial Setup Cost	Moderate to High	Low
Subscription Charges	Recurring	Not Required
Cloud Storage Cost	Required	Not Required
Maintenance Cost	Moderate	Minimal
Power Consumption	High (Smart Devices)	Low

4.1.3 Operational Feasibility

Operational feasibility is used to find out whether the system can work in the actual classroom settings. The proposed system is simple to use among both the instructors and the students. The students just have to tap an NFC tag to the option of MCQ that they have chosen. The Arduino automatically processes the response and the tag is detected by the RFID reader. The system does not involve highly technical training on how to use it by teachers. The offline design allows the continuous assessments with no network interruption. The system also has the capability of real-time assessment, whereby an instructor can evaluate the learning outcomes in real time as per the principles of OBE.

Table 4.3 Operational Feasibility Analysis

Operational Aspect	Existing Systems	Proposed System
Ease of Use	Requires Login & Device Setup	Simple Tag Selection
Technical Training	Required	Minimal
Dependency on Devices	High	None (No Smartphones)
Classroom Distraction	Possible	Eliminated
Real-Time Feedback	Yes	Yes (Offline Mode)

4.2 Software Requirement Specification

Software Requirement Specification (SRS) outlines the specific functional and non-functional requirements of the Classroom Assessment System (CAS) which is proposed on the basis of RFID/NFC. It is an official document specifying the operation of the system, the kind of data it is supposed to process, as well as the performance and security requirements of the system. The SRS allows the structure of system development to be consistent and geared towards the Outcome Based Education (OBE) principles.

4.2.1 Data Requirement

The system handles structured data that are created in classroom tests involving MCQ. As the proposed CAS is used in the offline mode, the entire information is stored locally in the

connected system database. These are the Unique Identifier (UID) of each NFC tag, the answer to the question (A/B/C/D) that the instructor chose, the question ID, the correct answer typed in by the instructor and the evaluation result. Moreover, timestamps can be used to ensure the right assessment tracking. The system should have the right data mapping between tag UID and selected option with consistency and integrity in the course of the evaluation. Data storage needs to be organized, safe as well as easy to access and analyse to enable reporting.

4.2.2 Functional Requirement

The functional requirements explain the key operations that the system should carry out in order to realize its objectives. The RFID reader should be able to read NFC tags and pass the UID to the Arduino board to be processed. Arduino is required to match the UID to a MCQ option which is already present in the database and save the answer in the local database. The system must enable the instructor to key in the right answer to each question after which automated analysis should then be done by comparing the responses stored to the correct response key. The system should provide a report on the outcomes such as the amount of right and wrong answers. All activities must be carried out in the real time without any internet availability so that classroom evaluation is maintained.

4.2.3 Performance Requirement

The system should provide effective and efficient performance in the classroom sessions. Response detection time must be as short as possible so that interactions are smooth and there will be no delays in assessments. The NFC tags should be read by the RFID reader at an acceptable range to avoid the occurrence of numerous inadvertent reads. The processing speed of the Arduino should be fast enough to accommodate real time recording and analysis of responses. The system must also be persistently available and functional during the lecture without failure so that there is 100 percent availability during assessment activities. Moreover, the system must be low consuming of power with the aim of energy efficiency.

4.2.4 Maintainability Requirement

The suggested CAS should have a simple maintenance and upgrading. Because the system is developed based on modular hardware systems including RFID reader, Arduino board and NFC tags, then different parts can be changed or even upgraded without any repercussions on the overall system architecture. The programmed arduino software is adaptable, and can be updated or changed in case of need, and can be changed however the user wishes further on in the future. Effective record keeping in the configuration and functional instructions of the system should be in place to ease the troubleshooting and maintenance. With the help of scalability, the system design should be incorporated in a manner that it can be scaled in the future to include more facilities like advanced analytics or graphical reporting modules.

4.2.5 Security Requirement

One of the vital factors of the proposed system is security. The CAS can be turned offline and local data storage means that it is exposed to external cyber threats to a significant extent. All the responses made by the students should be locked up in the local system to ensure that they are not accessed by any other unauthorized person. Response mapping is accurate and not tamperable based on the use of unique NFC tag UIDs. Only authorized instructors should be allowed to access the stored assessment data. The system removes cloud-based storage and reliance on internet resulting in greater confidentiality, integrity, and controlled access to academic records, which meets the criteria of secure educational assessment practices.

4.3 SDLC Model Used

The construction of RFID/NFC-based Classroom Assessment System (CAS) has been developed on the Waterfall Model. This model is linear and sequential in the development process as the various stages have to be done and then progressed to the next. It is best applied in a project where the requirements are clearly stipulated in the outset and are unlikely to be

altered too often. The Waterfall Model was viewed as suitable since the proposed CAS had clear Functional Requirements (FR) and Non-Functional Requirements (NFR), and was to be structured integration of hardware and software solutions.

The Classroom Assessment System incorporates the implementation of RFID reader, Arduino board, and NFC tags to support the offline mode of gathering responses based on MCQ. Since the system is not characterized by the constant change of requirements and has a well-defined development process, starting with the requirement analysis and ending with the deployment, the use of a sequential SDLC method guaranteed that the implementation was systematic. The Waterfall Model also assisted in ensuring that documentations, separation of phases, and controlled execution of all project life cycle phases was in place.

All the steps of the Waterfall Model are split into large steps:

Requirement Analysis Phase- The system requirements have been determined based on gaps in research that have been noted in the other online assessment systems. The requirement of an off-line, secure, cost effective, and distraction-free, classroom assessment mechanism had been determined. The SRS contained functional and performance requirements. The specifications of the hardware and software were made clear and thereafter to the next stage.

System Design Phase- It was at this stage that general system architecture was planned. The RFID reader, Arduino UNO, and NFC tags were chosen as the hardware components and they were conceptually interfaced. The information flow between tag detection, UID mapping, response storage and evaluation logic was designed. The database design and the evaluation mechanism were also scheduled to provide the integrity of the data and its confidentiality.

Implementation Phase- system development was done in this phase. Embedded C/C++ was used to program the Arduino to read the UID of NFC tags and match the UID with a predefined MCQ option. Evaluation logic of response was carried out and documents were

prepared and joined together and assembled hardware. The system has incorporated local data storage and automated results processing modules.

Testing Phase- The system developed was tested to ensure accuracy in RFID detection, response time and evaluation accuracy. Unit testing and integration testing was done to make sure that there was alignment between software and hardware components. During the testing, errors were found and rectified before actual deployment.

Deployment and Maintenance Phase- Once testing was done successfully the system was deployed in a classroom setting where it was utilized in real-time assessment. The future maintenance will include minor hardware modifications, Arduino code modification, and improving the system performance where required.

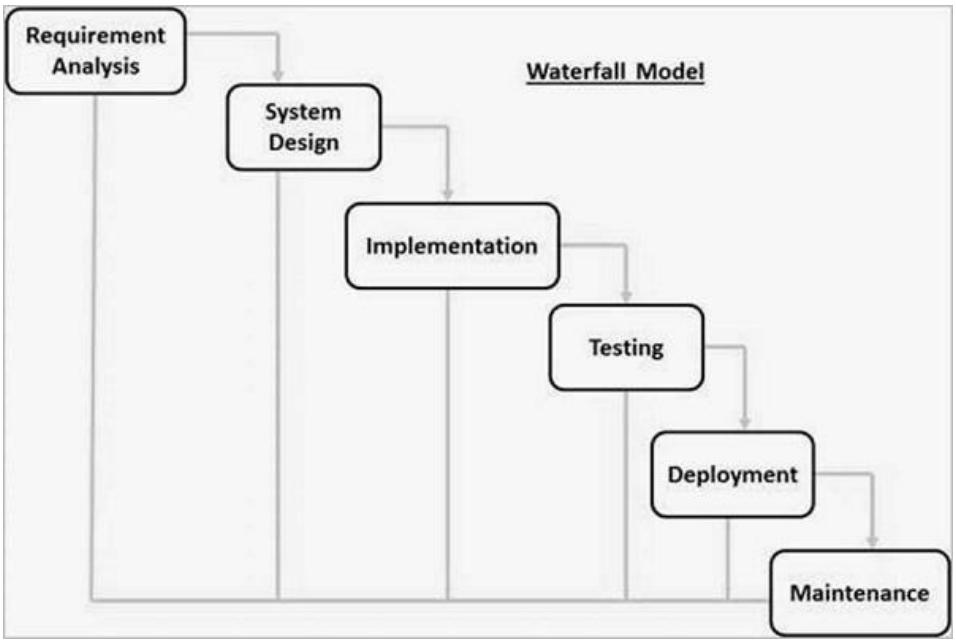

Figure 4. 1 View of Waterfall Model

4.4 System Design

The System Design stage offers a detailed plan of the proposed RFID/NFC-based Classroom Assessment System (CAS). It illustrates the structural layout, flow of data, communication methods, and logical operations of the system. During this stage, the functional and non-functional requirements outlined in the SRS are converted into an organized technical structure.

The design is such that it ensures the system works well when in offline mode while at the same time ensuring data confidentiality, response accuracy, and system reliability. The fact that the project combines hardware components such as the RFID reader, Arduino UNO, and NFC tags with software components such as the Node.js backend, MySQL database, and React.js frontend means that the design is modular and layered.

The system design requires basic operational functionality with minimal costs and capacity to expand while following Outcome-Based Education (OBE) standards. The project focuses on three main objectives which include reducing distractions and preventing unethical behavior and achieving complete internet independence.

4.4.1 Data Flow Diagrams

Data flow diagrams, or DFDs, basically show how information goes around in a system like the CAS. They are more about the logical steps of changing data, not the actual hardware stuff. It makes sense to look at it that way, from a higher level, to get the overall function.

In the CAS setup, everything starts with a student picking an option for the multiple choice question. They do this by scanning an NFC tag, I think. The RFID reader picks up the unique ID from that tag. Then it sends this ID over to the Arduino UNO using something called SPI protocol. The Arduino takes that ID and matches it up to one of the stored options in its memory.

After that processing, the response gets sent to a storage part. There, it ends up in a MySQL database, with extra details like the time it happened, which question it was, and who the student is. Once the teacher puts in the right answer key, the system compares everything. It looks at the stored answers against the correct ones and spits out some kind of report on performance.

The DFD Level 0 (Context Diagram) offers a high-level description in which the entire CAS system is modeled as a single process that interacts with two external entities: Student and Instructor. Students enter input in the form of scanned responses, whereas instructors enter answer keys and generate corresponding reports.

The DFD Level 1 breaks down the system into internal processes such as Tag Detection, UID Mapping, Response Validation, Data Storage, and Evaluation. This level of description offers a better insight into the step-by-step processing of data within the system. The structured data flow process ensures accuracy, eliminates the possibility of duplicate responses, and preserves the integrity of the system.

CHAPTER 5
IMPLEMENTATION

5.1 Introduction Tools and Technologies Used

Hardware-based MCQ Classroom Assessment Systems were developed to support real-time response collection, secure data processing, and effective classroom assessment management using Internet of Things (IoT) hardware, full-stack web technology, databases, and deployment tools. The design of the system is intended to minimize distraction by providing a structured assessment environment; and provide real-time feedback to students through seamless communication between the student's response device and the teacher's dashboard.

The core function of the system is based upon a microcontroller-based IoT Architecture that collects and sends responses from students to a central server. As opposed to traditional mobile applications, this system utilizes stand-alone hardware to reduce distraction, promote structure and increase the likelihood of student participation. The server processes received responses, validates them, determines grades and updates real-time performance analytics.
The Software Backend was written utilizing Node.js, which provides for rapid server side processing and RESTful API communication to the client. The Frontend Dashboard was created utilizing React.js with Bootstrap to provide a responsive, user friendly, and dynamic UI experience. The Teacher Dashboard enables instructors to develop MCQ's, view live responses, and produce performance reports.

Student Information, Question Banks, Response Logs, Timestamps, and Course Outcome Maps are stored within MySQL relational database management system. The system architecture is consistent with RESTful principles to facilitate structured communication between hardware modules, backend services, and the web UI.

Authentication methods such as JWT (JSON Web Tokens) and encryption protocols such as SSL/TLS may be utilized to provide secure access to the application and protect data. In addition, version control systems such as Git and GitHub will be used to manage source code and collaborative development. The application can be installed on a local server for institution-wide use or on a cloud platform to support multi-classroom deployments at scale.

Table 5.1 Tools and Technologies Used

Category	Tool / Technology	Purpose
Programming Language	JavaScript	Core development of frontend and backend components
Frontend Framework	React.js	Development of interactive teacher dashboard
UI Framework	Bootstrap	Responsive and user-friendly interface design
Backend Runtime	Node.js	Server-side logic, API handling, and real-time data processing
Database	MySQL	Storage of student records, question bank, responses, and analytics
IoT Hardware	Microcontroller (e.g., ESP8266 / ESP32 / Arduino)	Capturing and transmitting student responses
Communication Protocol	HTTP / RESTful API	Communication between hardware devices and backend server
Authentication	JWT (JSON Web Token)	Secure login and session management
Security	SSL / TLS	Encrypted data transmission

Category	Tool / Technology	Purpose
Version Control	Git / GitHub	Source code management and collaboration
Deployment Environment	Local Server / Cloud Platform	Hosting and deployment of the system
Development Tools	VS Code / Postman	Coding, API testing, and debugging

CHAPTER 6
TESTING AND MAINTENANCE

Testing and Maintenance stage will ensure Hardware-Based MCQ Classroom Assessment System to practice successfully in the real time classroom environment with high accuracy, security, and efficiency. As the system is a combination of IoT hardware and full-stack web technologies, extensive testing was necessary to ensure the work of both independent modules and the functioning of the entire system. Effective testing reduces failure of the system, integrity of data and boosts confidence by the users. On successful testing, maintenance strategies were established in order to provide reliability and scalability over time.

6.1 Testing

Functional, performance, and security requirements were tested at different levels in order to be satisfied within the system.

6.1.1 Unit Testing

Unit testing was done so as to test the functionality of each module individually. All the elements like authentication to the system on the part of a user, question authoring, response posting, API integration, and database storage were independently tested. This was to make sure that all modules do not fail to carry out their intended task until after the integration.

6.1.2 Integration Testing

Integration testing was aimed at communication verification between hardware, backend server, and frontend dashboard. The IoT device was run to ensure that the responses of students were sent to the server correctly and saved in the database. The real time data update and the dashboard visualization were also verified.

6.1.3 System Testing

System testing did test the entire workflow of the application. The overall system of teacher login, MCQ creation, submission of student response, evaluation, and report generation was under test as a single system. This made sure that there was no conflict in all modules.

6.1.4 Performance Testing

To measure performance in terms of the response time of the system and scalability, performance testing was performed. The interaction of several parallel replies was tested to determine whether the system was capable of managing the classroom level of participation without any lag or delay. The server and the database were checked so that they would not go out of proportion when being loaded.

6.1.5 Security Testing

Security testing was to protect against unauthorized access and breaches of data. Authentication systems were put through test to ensure the secure login. Endpoints of API were verified to avoid unauthorized manipulation of data. Access control to the database was also verified to ensure the data was confidential.

6.1.6 Test Cases

The following table summarizes the major test cases executed during the testing phase:

Table 6.1 Functional Test Cases

Test Case ID	Module Tested	Test Description	Expected Result	Status
TC01	Login Module	Valid teacher login credentials	Successful login	Passed
TC02	Login Module	Invalid login attempt	Error message displayed	Passed
TC03	Question Module	Teacher creates new MCQ	Question saved successfully	Passed
TC04	Response Module	Student submits answer	Response stored in database	Passed
TC05	Real-Time Update	Multiple students respond simultaneously	Dashboard updates instantly	Passed
TC06	Report Module	Generate performance report	Accurate result displayed	Passed

Table 6.2 Performance & Security Test Cases

Test Case ID	Category	Test Description	Expected Outcome	Status
TC07	Performance	50+ responses at same time	No system delay	Passed
TC08	Database	Verify response storage	Correct data saved	Passed
TC09	API Testing	Validate response transmission	Proper server communication	Passed
TC10	Security	Unauthorized API access	Access denied	Passed

6.2 Maintenance

Maintenance facilitates the performance, reliability and flexibility of the system once it is deployed. Since the suggested system might be implemented on the routine basis within the classroom environment, it is essential to revise and check it on a periodic basis.

6.2.1 Corrective Maintenance

Corrective maintenance involves identification and resolution of errors or bugs that have been detected during the post deployment. This entails troubleshooting hardware communication issues, database issues or front end display issues.

6.2.2 Adaptive Maintenance

Adaptive maintenance is performed in order to change the system according to the change in the institutional requirement or the technological progress. Such a manner of new authentication implementation, server configuration optimisation, or dashboard functionality.

6.2.3 Perfective Maintenance

Perfective maintenance is also geared towards improving system usability and performance. It can be enhanced as per the feedback of the educators by including more analytics components, enhanced interface, or increased reporting features.

6.2.4 Preventive Maintenance

Preventive maintenance will include regular check up of the system, backing of database and updating of security to prevent cases of failure. Periodic monitoring ensures that hardware equipment is operating in the desired manner and that server performance is not on the decline.

Table 6.3 Maintenance Activities

Maintenance Type	Activity	Purpose
Corrective	Bug fixing and error resolution	Ensure smooth operation
Adaptive	Technology and feature upgrades	Maintain compatibility
Perfective	UI/UX improvements and analytics enhancement	Improve user experience
Preventive	Regular backups and security updates	Prevent system failures

CHAPTER 7
RESULTS AND DISCUSSIONS

The Hardware-Based MCQ Classroom Assessment System was implemented and demonstrated the successful real-time data collection and correct data processing. The IoT device was able to transmit student responses without errors to the database which was secured and stored. The dashboard provided to the teachers was updated immediately, which allowed rapid assessment and tracking of performance.

The system enhanced classroom interaction and minimized distraction by mobile. The calculation of scores automatically reduced manual labor on the assignment and provided clarity of evaluation. It was tested in performance conditions and the capacity to stabilize even under several simultaneous responses was seen to be stable.

7.1 Description of Modules

The Hardware-Based MCQ Classroom Assessment System is made up of various integrated modules which operate in concert with each other to provide a seamless classroom assessment and real-time assessment. The modules have been developed to carry out a particular role whilst ensuring a smooth interaction with the other elements of the system.

Authentication Module is in charge of the secure access of teachers and administrators. It checks the credentials of users through validation of credentials with the back end and only the authorized user can view the dashboard and assessment controls. The module is able to control sessions in an efficient manner and avoid illegal access to the system.

Question Management Module enables instructors to design, update and control the multiple choices. The questions in the database are in the structured form of questions with their options and correct answers. By utilizing this module, it will be possible to manage the

assessment system in an organized way and guarantee that the questions could be connected to particular subjects or learning outcomes.

The Hardware Response Module provides the students with an opportunity to submit the answers with the help of the special IoT-based hardware device. All devices are singly identified and linked to the central server. As a student uses the hardware buttons to make a choice the reply is sent to the backend server immediately and stored in the database. This module will remove distractions in the classroom since there will be no Smartphones that will be used.

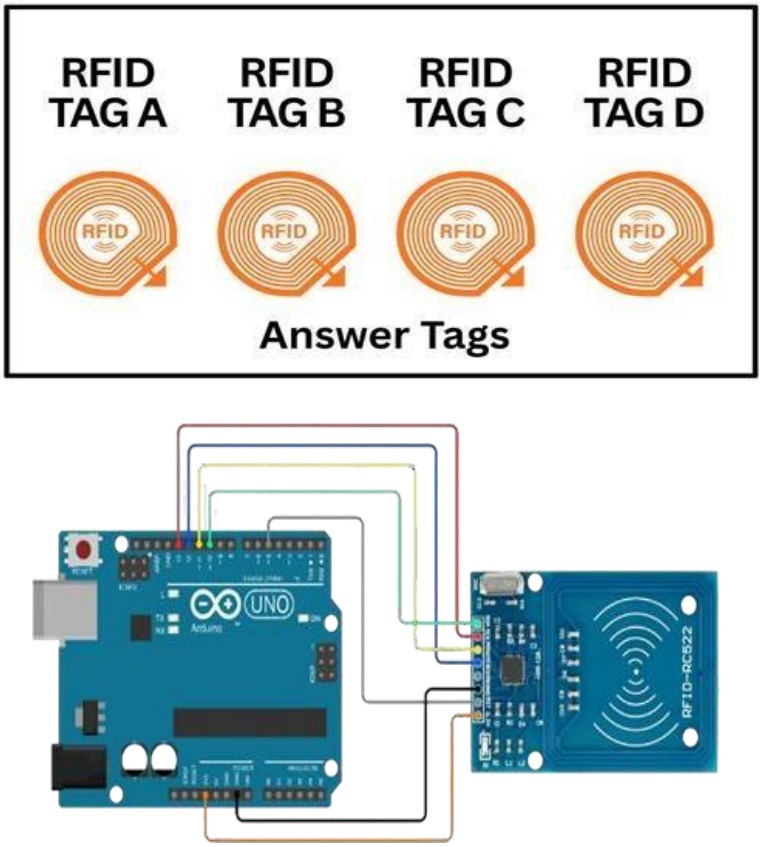

Figure 7.1 RFID Tags and Arduino Board Representation

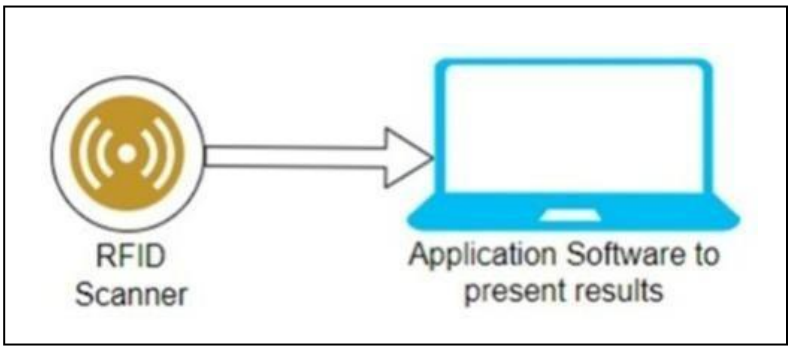

Figure 7.2 Graphical representation of the circuit

The Real-Time Monitoring Dashboard enables the teacher to have real-time information on student participation and the ability to answer correctly. The results of the answers are shown on the dashboard immediately after they have been submitted, which enables instructors to analyze the performance of the classroom immediately. This aspect allows rapid teaching change according to student comprehension.

The Generation of the Report automatically scores the responses and evaluates them. It produces performance reports in detail, both at the individual level and the overall level of the whole classroom. The reports assist the teachers to monitor academic progression and where the students need more assistance.

7.2 Key Findings of the Project

The proposed system implementation and evaluation brought important results. The system was able to decrease the reliance on the mobile devices which minimized distractions during classroom assessments. It improved the level of student engagement and participation since students could engage actively in the process of their assessment, which was made possible by the real-time response collection mechanism.

Automatic evaluation service played a significant role in reducing the manual checking of the work conducted by the teachers. The communication provided by the IoT was stable and quick

in the transfer of responses in cases where more than two devices were made at the same time. The system was scalable and was stable to run with many people using it at the same time.

Security testing was done to ensure that unauthorized access requests could be blocked by means of authentication. Moreover, educators could discover the poor learning areas within a short time frame on the basis of instant analytics on the dashboard. In general, the system was efficient, reliable, and corresponded with the principles of Outcome-Based Education (OBE) since the results of the assessments were measurable and structured.

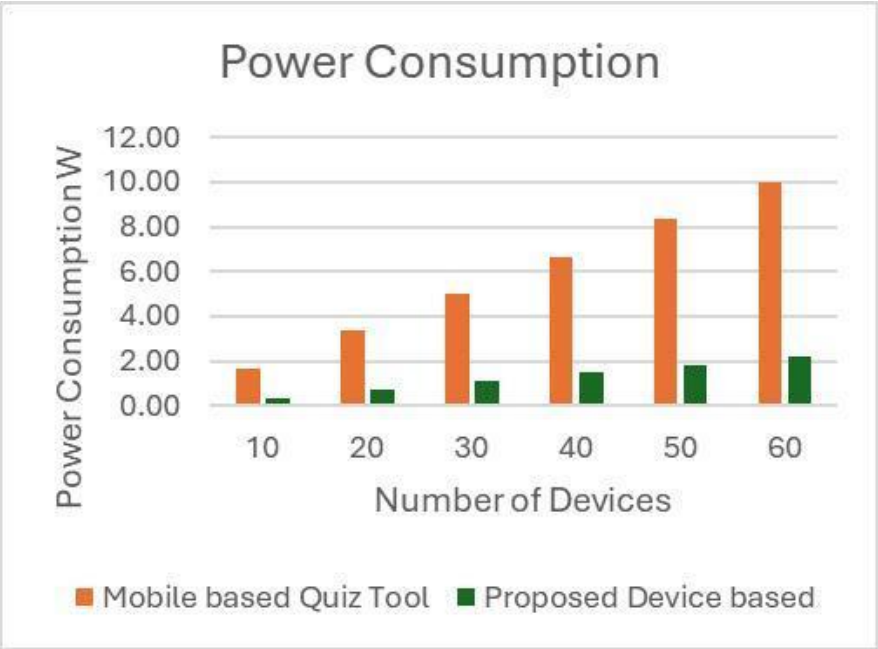

Figure 7.3 Energy Efficiency Evaluation of Proposed Hardware-Based Assessment System

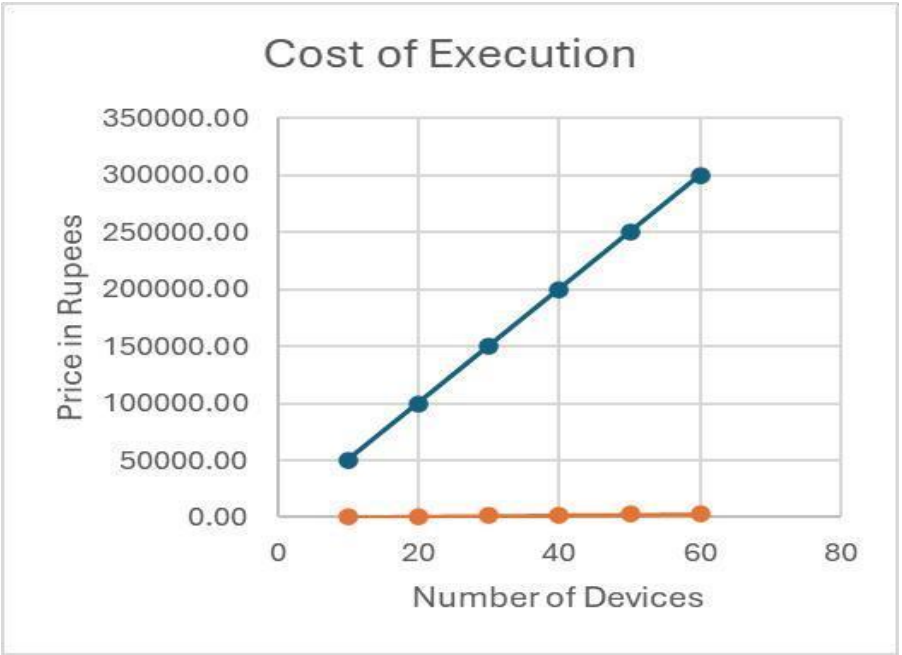

Figure 7.4 Cost Evaluation of Mobile-Based Quiz Tool versus Proposed Hardware-Based System

7.3 Brief Description of Database

The system is developed based on MySQL database where all the necessary data, such as user credentials, questions, answers, and performance history, are stored. The database design guarantees the data integrity, consistency, and efficient access of information.

Users table contains authentication information about teachers and administrators. Questions table holds the MCQ information, question text, the possible available options and the correct answers. Each selected option of the students is logged into the Responses table with the help of timestamps and device identification. The Results table contains calculated scores and the general performance indicators based on the response information.

The database design is a relational design and a teacher can create various questions and each question can receive various responses depending on the device of the student. Table association guarantees correct mapping of responses to certain questions and students. A good

indexing and schema design will increase performance of queries and make dashboard updates an almost real-time process.

The database was well tested to ensure that it was accurate and consistent. All the data was stored properly without any duplication and loss and retrieval operations were done efficiently which proved that the backend architecture of the system was reliable.

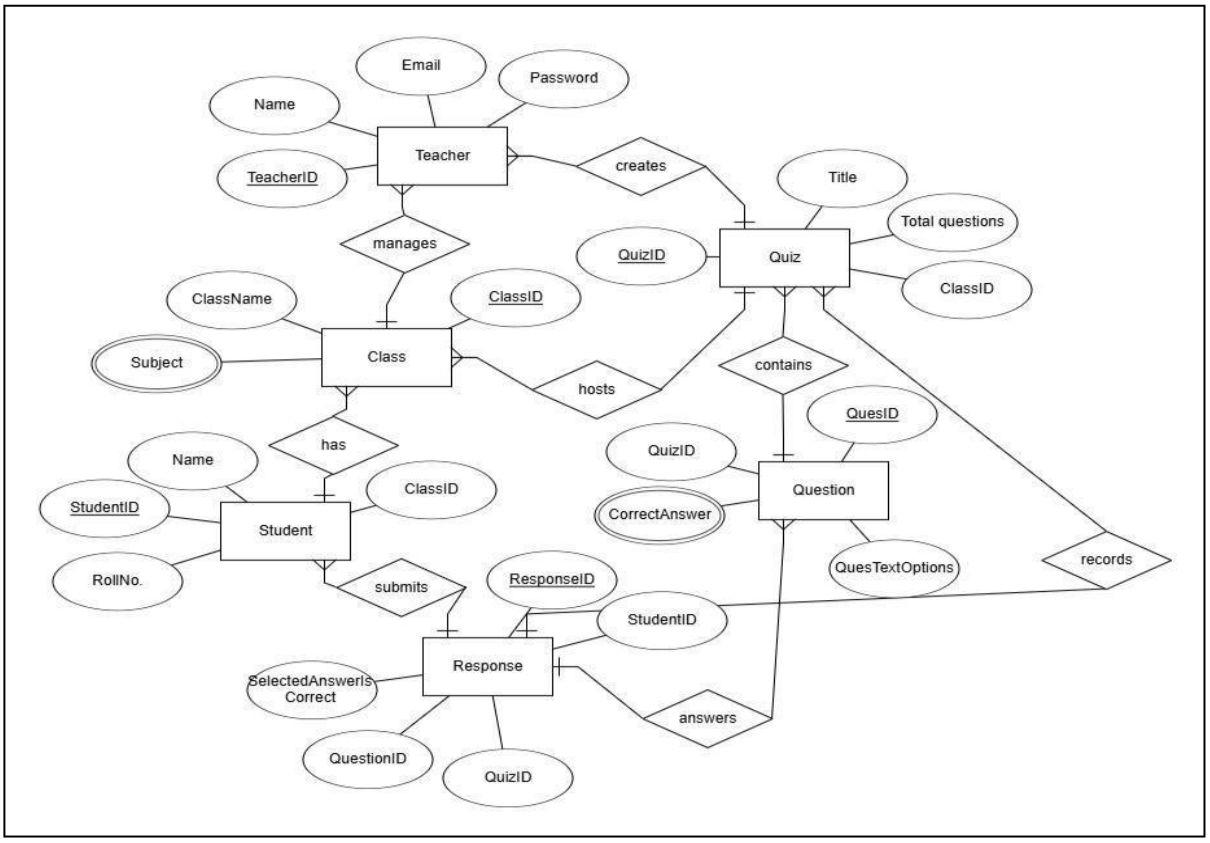

Figure 7.5 ER Diagram

CHAPTER 8
CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

The development of the Hardware-Based MCQ Classroom Assessment System with the introduction of the Outcome-Based Education (OBE) Integration can be seen as a milestone in the change of the approach to classroom assessment. The project was initiated to overcome the weakness of the paper-based testing and the smartphone-reliant quiz websites which specifically involved delayed feedback and unstructured analytics, as well as Smartphone distractions in the classroom. The system will provide a disciplined and active participation through the introduction of a dedicated hardware device to the interaction of student responses creating a focused and controlled assessment environment.

Communication between student hardware and the central server can be easily made possible through the integration of IoT-enabled hardware with a web-based dashboard. React.js, Node.js, and MySQL were used as a frontend, a backend processing engine, and a database management system, respectively, which guarantees the stable performance, real-time data processing, and scalability. The system allows the teachers to design and administer MCQ-based assessment effectively and automatically store the responses and process analytical feedback. This saves on the need to work hard and enhances the accuracy and transparency of evaluation.

The ability of the proposed system to comply with the principles of Outcome-Based Education (OBE) is also its strength. The system aids in tracking measurable academic performance by mapping assessments with certain course outcomes and learning objectives to track them. Real time analytics assist educators to determine areas of weaknesses, track student progress and implement evidence-based teaching methods. The project is strengthening institutional academic monitoring mechanisms in addition to improving engagement.

Overall, the system successfully demonstrates how the combination of embedded hardware and contemporary web technologies could be used to turn traditional practices in the classroom into an intelligent, engaging, and performance-based learning environment. The solution is affordable, flexible, and can be disparate in schools, colleges, and universities that are in need of technology-based assessment structures.

8.2 Future Scope

One of the strengths of the proposed system is that it is based on the principles of the Outcome-Based Education (OBE). The system aids in the tracking of academic performance by mapping the assessment to the particular outcome of a course and the learning objectives. Real-time analytics would help teachers to understand the field of weakness, monitor the development of the students, and accept the data-driven methods of teaching. The project does not only increase the engagement but also makes the institutional academic monitoring mechanisms stronger.

In general, the system clearly proves that the combination of embedded hardware and latest web technologies is capable of redefining the traditional classroom processes into a smart, interactive, and result-oriented learning ecosystem. The solution is affordable, flexible, and can be adapted to schools, colleges, and universities that aim in adopting technology-driven assessment models though as it is currently implemented, there are quite a number of opportunities that can be developed and extended. The adoption of Artificial Intelligence (AI) and Machine Learning (ML) algorithms to conduct advanced learning analytics is also one of the key areas of improvement. History performance data could be studied using predictive models to predict trends, at-risk students, and offer learning suggestions to students individually. This would make it a response-gathering tool to a smart academic decision-support system.

The system can be additionally configured to accommodate a variety of different assessment types, such as descriptive answers, tests based on short coding, adaptive quizzes, and individually specific evaluation modules. It would be possible to integrate with the existing

Learning Management Systems (LMS) and synchronize course-related materials, attendance, and assessment data seamlessly. Scalability would also be extended by the possibility to use cloud-based deployment in order to have two or more classrooms and multiple institutions with a centralized administration structure.

Moreover, the hardware module can be advanced to support secure authentication protocols like RFID based authentication or biometric authentication to avoid impersonation and secure participation. Implementing a mobile monitoring application to the administrators and parents would enhance transparency and communication of the student progress. Performance badges and leaderboards can also be incorporated as additional motivation and engagement tools.

As the world continues to progress in terms of technology, the proposed system can also be developed to establish a holistic smart classroom environment that will help support mass-based digital education. The future developments will not only be an improvement but will also help move towards an innovation-driven educational infrastructure.

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1 message

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Comments to Author(s)
The paper presents a practical and innovative hardware-based approach for classroom assessment using RFID/NFC technology, addressing key challenges associated with smartphone- and internet-based quiz systems such as distraction, connectivity dependency, privacy concerns, and cost. The focus on Outcome Based Education (OBE) and real-time lecture assessment enhances the practical relevance of the work.

The proposed methodology is clearly explained, and the comparison with existing online quiz tools highlights the advantages of the offline, low-cost solution. The discussion on power consumption and cost analysis further strengthens the applicability of the device, particularly for institutions with limited infrastructure.

Minor improvements could include a more detailed quantitative evaluation with larger classroom deployments and performance metrics such as response accuracy and scalability. Overall, the work makes a useful contribution to educational technology and classroom assessment methods.

Recommendation
Accept with Minor Revision

Regards
Team International Conference on Innovative Practices in Technology and Management (2026)

Research Paper (If Presented/Published)

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CERTIFICATE OF PARTICIPATION IN RESEARCH PAPER PRESENTATION

This is to acknowledge the participation of Lakshan (Academician/Corporate/Research Scholar/Student) for the **Research Paper Presentation** in the **International Conference on Innovative Practices in Technology and Management, (ICIPTM – 2026)**, on 19th-20th February 2026, held at Amity University, Noida.

The paper titled A tool for lectures assesment as per OBE policy. is worth appreciating for its efforts in generating, analysing and presenting the research content.

Prof. (Dr.) Gurinder Singh
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Proof of patent publication

(12) PATENT APPLICATION PUBLICATION

(19) INDIA

(21) Application No.202511050354 A

(22) Date of filing of Application :26/05/2025

(43) Publication Date : 13/06/2025

(54) Title of the invention : A TOOL FOR LECTURE ASSESSMENT AS PER OBE POLICY

(51) International classification	:G06Q0050200000, G09B0007020000, G09B0005060000, G09B0007000000, G09B0005020000
(86) International Application No Filing Date	:NA :NA
(87) International Publication No	: NA
(61) Patent of Addition to Application Number Filing Date	:NA :NA
(62) Divisional to Application Number Filing Date	:NA :NA

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(57) Abstract :

The present invention relates to a hardware-based response system designed for real-time classroom assessments, aligning with Outcome-Based Education (OBE) principles. The system enables teachers to conduct MCQ-based evaluations without mobile phones, thereby reducing distractions and ensuring focused student participation. It comprises a hardware device for student responses and an integrated software platform for assessment, feedback, and performance tracking. The tool provides immediate feedback, helping educators measure learning outcomes effectively. By automating assessment processes, this invention enhances student engagement, improves instructional strategies, and ensures efficient performance evaluation in educational settings.

No. of Pages : 14 No. of Claims : 4

Project Report Plagiarism Report

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